

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

PRINCIPAL COMPONENT ANALYSIS

SANJOY DASGUPTA, PROFESSOR

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE



Projection onto Multiple Directions

Projecting $x \in \mathbb{R}^d$ into the k —dimensional subspace defined by vectors $u_1, \dots, u_k \in \mathbb{R}^d$.

Projection onto Multiple Directions

Projecting $x \in \mathbb{R}^d$ into the k –dimensional subspace defined by vectors $u_1, \dots, u_k \in \mathbb{R}^d$.

This is easiest when the u_i 's are **orthonormal**:

- They have length one.
- They are at right angles to each other: $u_i \cdot u_j = 0$ when $i \neq j$

The projection is a k -dimensional vector:

$$(x \cdot u_1, x \cdot u_2, \dots, x \cdot u_k) = \underbrace{\begin{pmatrix} \longleftrightarrow u_1 \longrightarrow \\ \longleftrightarrow u_2 \longrightarrow \\ \vdots \\ \longleftrightarrow u_k \longrightarrow \end{pmatrix}}_{\text{call this } U^T} \begin{pmatrix} \updownarrow \\ x \\ \updownarrow \end{pmatrix}$$

U is the $d \times k$ matrix with columns $u_1, \dots, u_k$.

The Best Direction

Let Σ be the $d \times d$ covariance matrix of X .

In $O(d^3)$ time, we can compute its **eigendecomposition**, consisting of

- real **eigenvalues** $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d$
- corresponding **eigenvectors** $u_1, \dots, u_d \in \mathbb{R}^d$ that are orthonormal (unit length and at right angles to each other)

Fact: Suppose we want to map data $X \in \mathbb{R}^d$ to just k dimensions, while capturing as much of the variance of X as possible. The best choice of projection is:

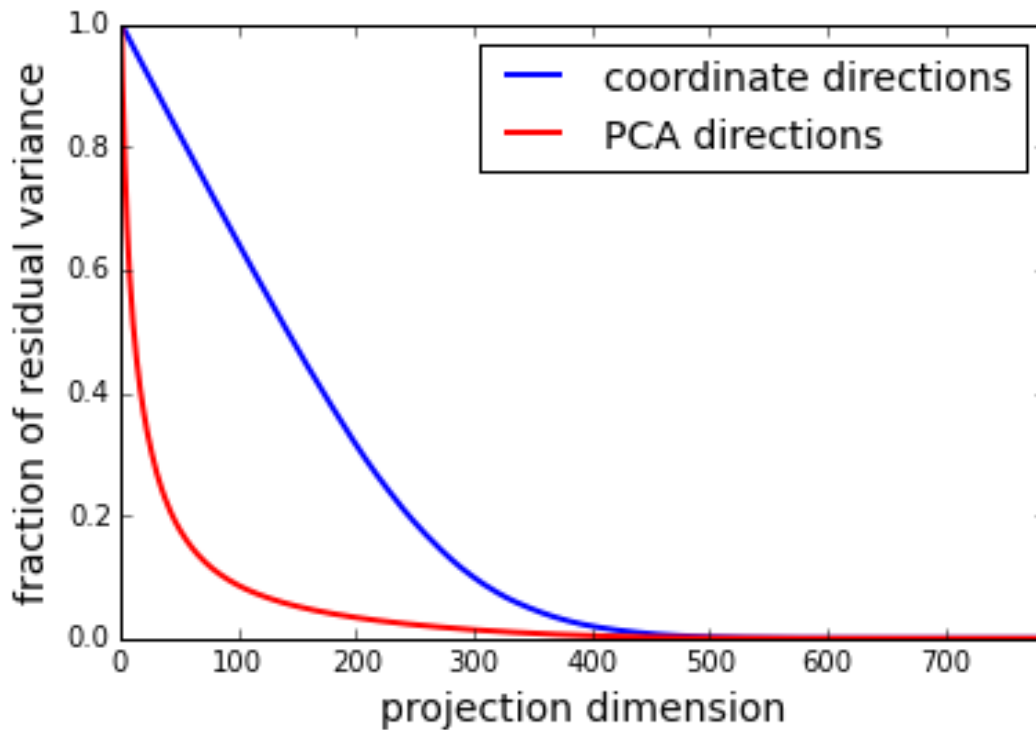
$$x \mapsto (u_1 \cdot x, u_2 \cdot x, \dots, u_k \cdot x),$$

where u_i are the eigenvectors described above.

This projection is called **principal component analysis** (PCA).

Example: MNIST

Contrast coordinate projections with PCA:



Applying PCA to MNIST: Examples



Reconstruct this original image from its PCA projection to k dimensions.

$k = 200$



$k = 150$



$k = 100$

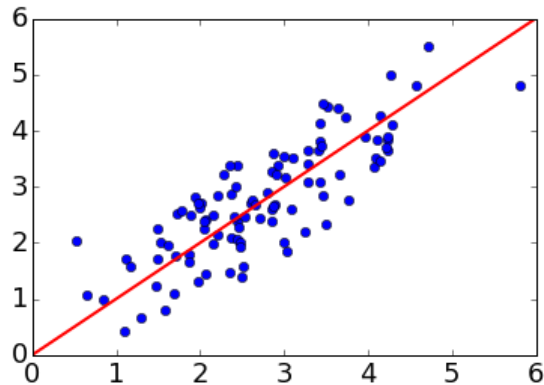


$k = 50$

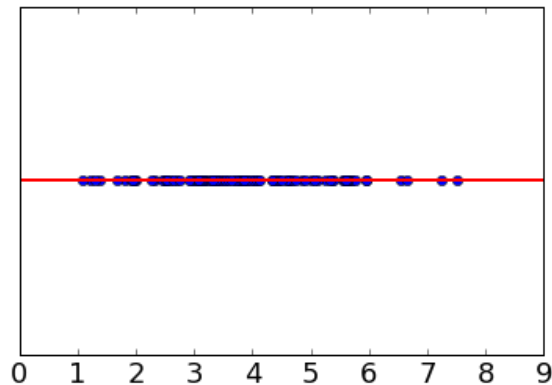


How do we get these **reconstructions**?

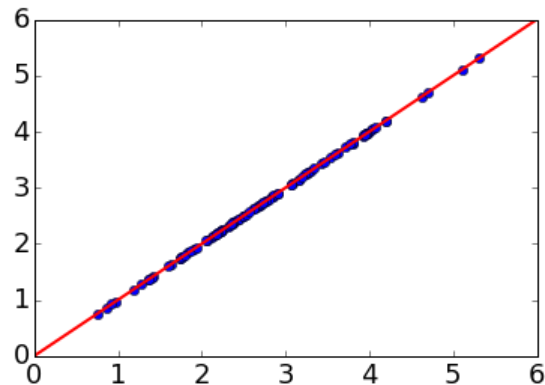
Reconstruction from a 1-d Projection



Projection onto $\mathbb{R}$:



Reconstruction in $\mathbb{R}^2$:



Projection onto Multiple Directions

Projecting into the k -dimensional subspace defined by **orthonormal** $u_1, \dots, u_k \in \mathbb{R}^d$.

The projection is a k -dimensional vector:

$$(x \cdot u_1, x \cdot u_2, \dots, x \cdot u_k) = \underbrace{\begin{pmatrix} \longleftrightarrow u_1 \longrightarrow \\ \longleftrightarrow u_2 \longrightarrow \\ \vdots \\ \longleftrightarrow u_k \longrightarrow \end{pmatrix}}_{\text{call this } U^T} \begin{pmatrix} \updownarrow \\ x \\ \updownarrow \end{pmatrix}$$

The reconstruction from this projection is:

$$(x \cdot u_1)u_1 + (x \cdot u_2)u_2 + \dots + (x \cdot u_k)u_k = UU^T x.$$

MNIST: Image Reconstruction



Reconstruct this original image x from its PCA projection to k dimensions.

$k = 200$



$k = 150$



$k = 100$



$k = 50$



Reconstruction $UU^T x$, where U 's columns are top k eigenvectors of Σ .